



Lightweight and Robust Partially Concealed Human Detection in Cluttered Scenes Using Elastic Shape Analysis

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OUTLINE

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2. Example imagery and contour extraction
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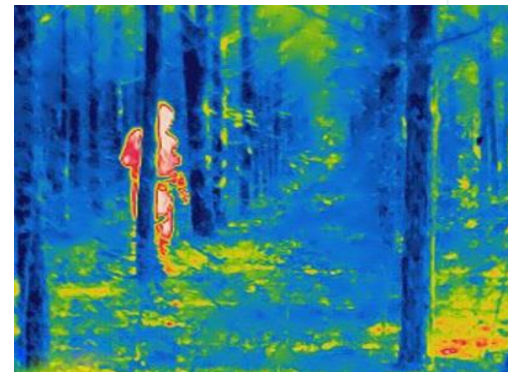
Elastic Shape Analysis (ESA) in a Nutshell

1



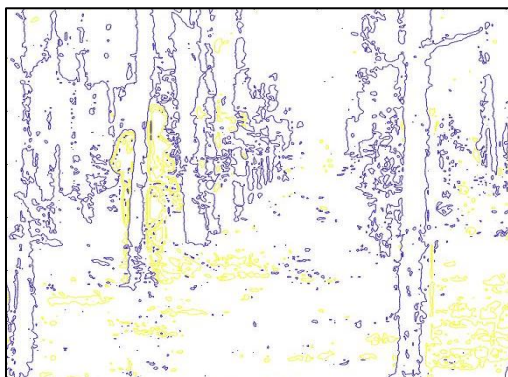
Training Set

2



IR Imagery (found at [1])

3



Contour Extraction

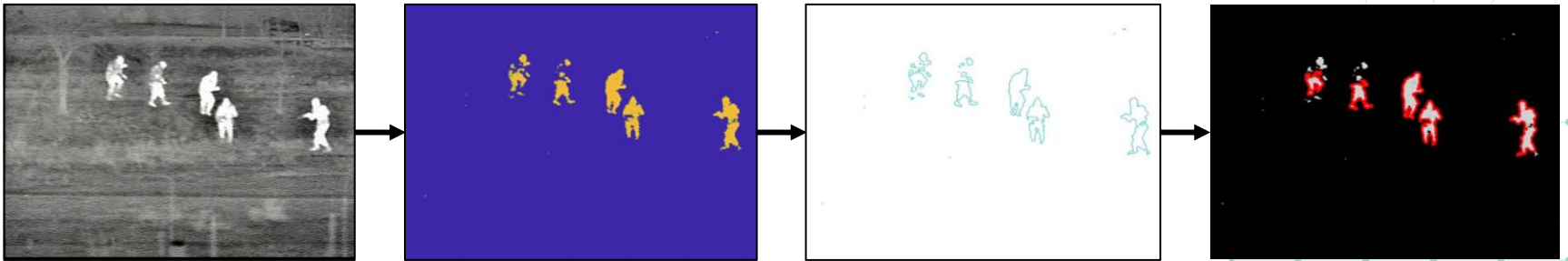
4



Target Detection

Why Do We Care About ESA

Remember: There's no free lunch in machine learning!



Original IR image may be found at [2]

Allow for low-size weight and power (low-SWaP) due to a tiny memory requirement for training.

It also mitigates over-parameterization due to consideration of only geometric invariants rather than emphasizing specific features.

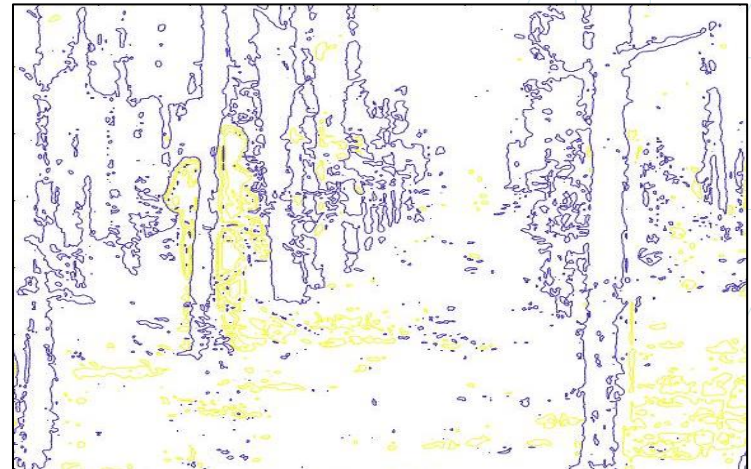
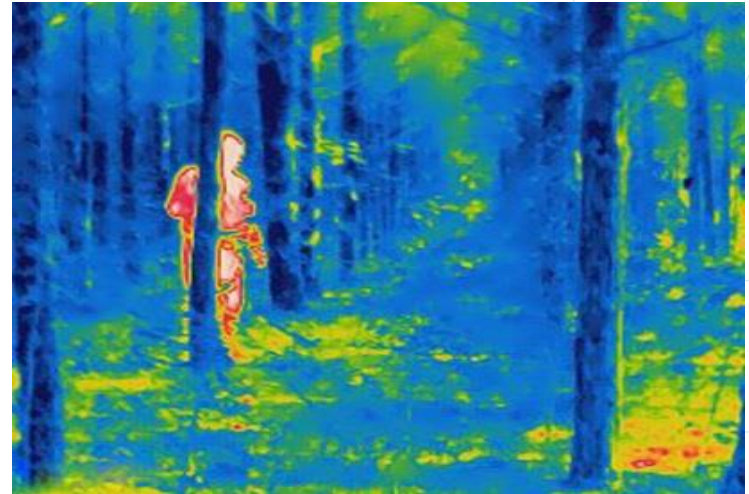
Example Imagery and Contour Extraction

Our goal is to show ESA's effectiveness at detecting/classifying- humans in a cluttered forest scene using an algorithm found in [3] – this algorithm is known as the Path Straightening Algorithm (PSA)

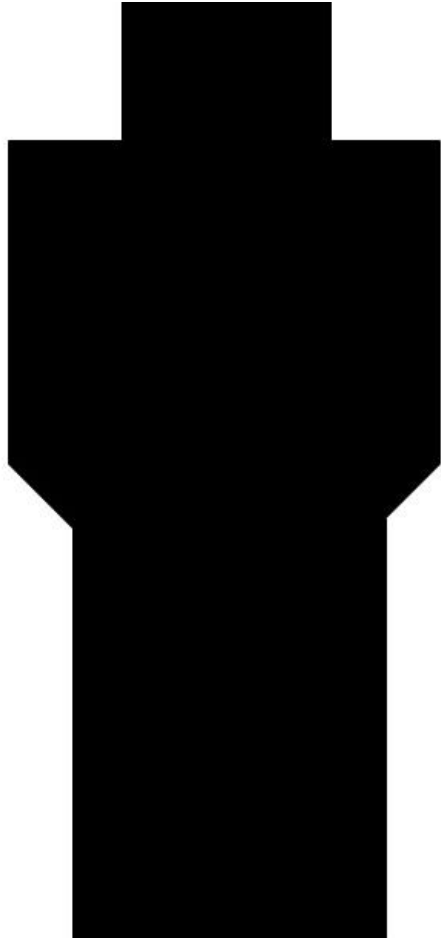
Candidate contours are extracted using MATLAB's *isocontour* function.

The true range and IFOV were not available, so maximum and minimum areas of a human are thus estimated based on reference data.

ESA scores focus attention on human-like figures due to their geometric invariants and their expected size within the scene – this information leads to the development of the APSA or Augmented Path Straightening Algorithm.



Training ESA to Detect Humans



The reference silhouette considered is given here.

The silhouette is necessarily an abstract representation of a human because we are only interested in the invariant portions of what constitutes a human. For example, we care about the location of head relative to the shoulders and legs.

This reference is applied sequentially through ESA such that the comparison score is made between the reference and the extracted candidate contour.

The best scoring contour is thus our human.

Results of Applying ESA to Detect a Human in a Forest Scene

Given the contours present in the image, our ESA algorithm will choose the contour that most closely matches the silhouette.

This is done based on examining the geometric invariants present and minimizing an energy functional meant to find the easiest path connecting the reference contour to the test contour.



Distinguishing between (Un)Manned Boats for Navy Applications



Original IR image may be found at [4]



Original IR image may be found at [5]



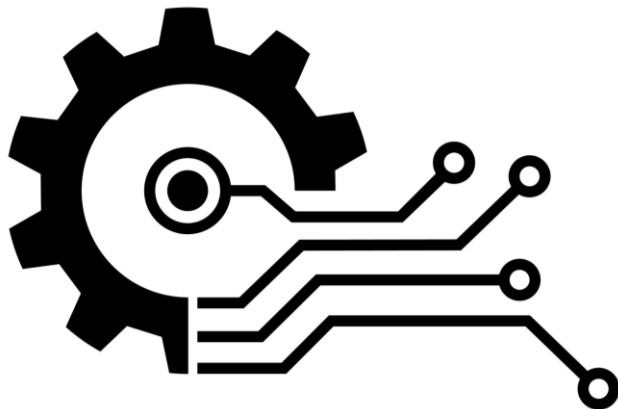
Original image may be found at [6]

Distinguishing between manned/unmanned boats is of great importance to our Navy's ability to engage with a potential aggressor.

There are certain key features present that are "tells" that classify whether a boat is manned or unmanned.

ESA allows us to find these "tells" and thus distinguish between a manned or unmanned boat – thus ensuring fast engagement policy determination.

Measurement of Computational Resources Consumed for low-SWAP Applications



Note that ESA operates on a single-look basis and the only “training” that it requires is the use of a (set) of reference contours.

This fact makes it light-weight in terms of memory consumption; our reference contour only requires 100 double precision numbers or 0.8 kilobytes.

The time consumed for ESA-based pattern classification running on MATLAB is 0.6 seconds.

It is expected that a C++ instantiation of this algorithm would require less time.

Key Takeaways and Closing Thoughts

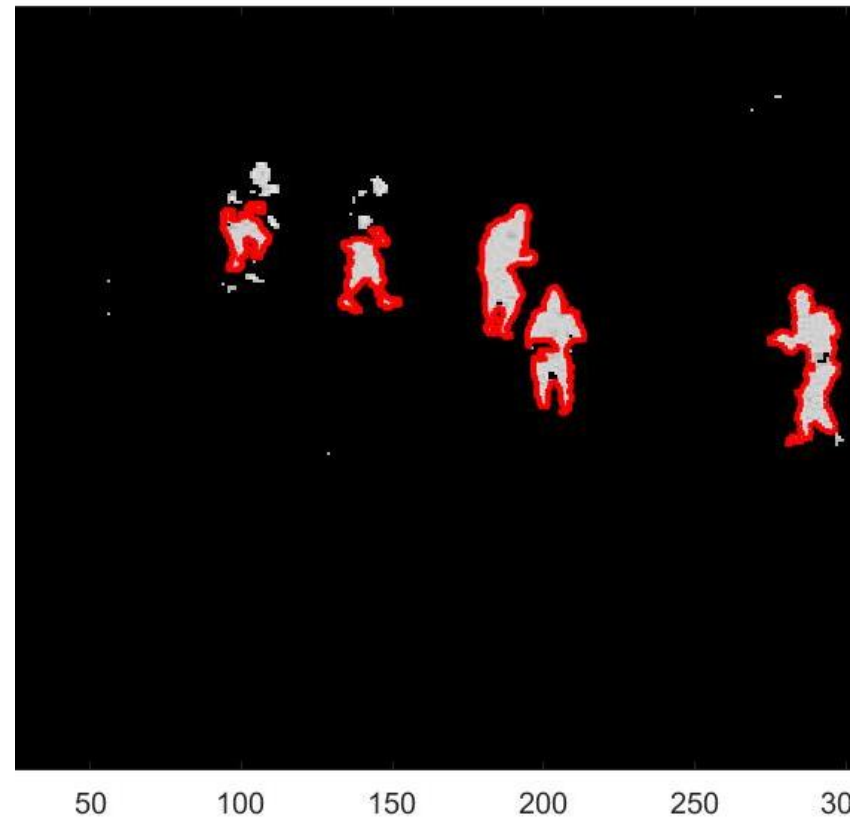
Classification performance relies upon trust in the contour-extractor to provide contours which may be matched against using ESA.

The curve extractor can be confused by human-like shadows when using electro-optic imagery. Hence there is an emphasis on IR imagery, but this is not a strict requirement.

Quick scene-level reasoning is possible with minimal a priori data. Incomplete representations of a human are a well-known issue in the context of autonomous vehicles and can lead to an enormous liability when using a conventional deep learning model [7].

We propose the use of shape analysis as a kind of “referee” that can offer further distinctions in the classification of a particular object.

Misclassification can be avoided by using a Bayesian classifier to better distinguish between human/non-human patterns.



References

- [1] Image found at: [Directed Infrared Countermeasure \(DIRCM\) – EMSOPEDIA](#)
- [2] Image found at: [Ronin Tactical's New Site: Raytheon L-3 Thermal-Eye MX-1 \(X-100xp\) Thermal Imaging Camera](#)
- [3] A. Srivastava and E. Klassen, Functional and Shape Data Analysis, Springer-Verlag, New York, 2016.
- [4] Image found at: [Abandoned cargo ship damaged in Houthi strike to be towed from Red Sea to Djibouti within days | The Times of Israel](#)
- [5] Image found at: [China, Russia and Iran to hold joint naval drills in Gulf of Oman \(axios.com\)](#)
- [6] Image found at: ['Ukrainian suicide drone' washes up on beach near Russian base after evading patrols - World News - Mirror Online](#)
- [7] GM's Cruise Halts All US Robotaxi Service After Suspension for Pedestrian Who Was Dragged, [GM's Cruise Halts All US Robotaxi Service After Suspension for Pedestrian Who Was Dragged | WIRED](#)